1. The One-Dimensional Quantum Spin Chain

Among many-body quantum systems, the one-dimensional occupies a special position. Owing to strong quantum fluctuations, the system is a quantum liquid, and down to zero temperature, no ordered state emerges. In this chapter, the one-dimensional quantum spin system will be discussed, being the introduction to the theory of strongly correlated systems.

1.1 The $S = 1/2$ XXZ Spin Chain

First, we consider the following Hamiltonian:

$$H = J_\perp \sum_i \left(S_i^x S_{i+1}^x + S_i^y S_{i+1}^y\right) + J_z \sum_i S_i^z S_{i+1}^z \,. \tag{1.1.1}$$

Here, $\boldsymbol{S}_i$ is the spin $S = 1/2$ operator. Expressing the $\uparrow$ spin state as $[1,0]^t$ and the $\downarrow$ state as $[0,1]^t$, and using the Pauli matrices $\boldsymbol{\sigma} = (\sigma^x, \sigma^y, \sigma^z)$

$$\sigma^x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \sigma^y = \begin{bmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{bmatrix}, \qquad \sigma^z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix},$$

the spin operator is given by $\boldsymbol{S}_i = (\hbar/2)\boldsymbol{\sigma}_i$. The spin operator satisfies

$$\left[S_i^\alpha, S_j^\beta\right] = \mathrm{i}\hbar\delta_{ij}\varepsilon_{\alpha\beta\gamma}S_i^\gamma \,. \tag{1.1.2}$$

$\varepsilon_{\alpha\beta\gamma}$ is the total antisymmetric tensor with $\varepsilon_{xyz} = 1$. In what follows, we usually set $\hbar = 1$. $J_\perp$ as well as J_z are the nearest-neighbour exchange interactions; and for the case $J_\perp = J_z$, the Hamiltonian is called the Heisenberg model.

From the commutation relation (1.1.2), it is clear that the unitary operator

$$U_i(\boldsymbol{n}, \theta) = \mathrm{e}^{\mathrm{i}\theta \boldsymbol{n}\cdot\boldsymbol{S}_i} \tag{1.1.3}$$

rotates the spin. Here, $\boldsymbol{n}$ is the unit vector, and θ is the rotation angle. For example, the case where $\boldsymbol{n} = \boldsymbol{e}_z = (0,0,1)$,

$$U_i(\boldsymbol{e}_z, \theta) S_i^\alpha U_i^\dagger(\boldsymbol{e}_z, \theta) = \tilde{S}_i^\alpha(\theta) \,, \tag{1.1.4}$$

leads to

$$\frac{\mathrm{d}\tilde{S}_i^x(\theta)}{\mathrm{d}\theta} = U_i \cdot \mathrm{i}\,[S_i^z, S_i^x]\,U_i^\dagger = -\tilde{S}_i^y(\theta)\,, \tag{1.1.5a}$$

$$\frac{\mathrm{d}\tilde{S}_i^y(\theta)}{\mathrm{d}\theta} = \tilde{S}_i^x(\theta)\,, \tag{1.1.5b}$$

$$\frac{\mathrm{d}\tilde{S}_i^z(\theta)}{\mathrm{d}\theta} = 0\,. \tag{1.1.5c}$$

solution is then given by

$$\tilde{S}_i^x(\theta) = S_i^x \cos\theta - S_i^y \sin\theta\,, \tag{1.1.6a}$$

$$\tilde{S}_i^y(\theta) = S_i^x \sin\theta + S_i^y \cos\theta\,, \tag{1.1.6b}$$

$$\tilde{S}_i^z(\theta) = S_i^z\,. \tag{1.1.6c}$$

ously, this describes a rotation around the z-axis. Next, constructing the ry operator $T = \prod_{n:\mathrm{odd}} U_n(\boldsymbol{e}_z, \pi)$ from U_i, because of

$$TS_i^x T^\dagger = (-1)^i S_i^x\,, \tag{1.1.7a}$$

$$TS_i^y T^\dagger = (-1)^i S_i^y\,, \tag{1.1.7b}$$

$$TS_i^z T^\dagger = S_i^z\,. \tag{1.1.7c}$$

ransformation $THT^\dagger$ of (1.1.1) interchanges $J_\perp \to -J_\perp, J_z \to J_z$. We ude that the sign of $J_\perp$ is not essential.

n the other hand, the sign of J_z plays an essential role for the quantum m. This is owing to the fact that, different from (1.1.7), the commutation on (1.1.2) of the spin components is changed for $\boldsymbol{S}_i \to -\boldsymbol{S}_i$. First, we ruct with S^x and S^y:

$$S^\pm = S^x \pm \mathrm{i}S^y\,. \tag{1.1.8}$$

ıse of

$$S^+ = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad S^- = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

the operator that flips the spin $\uparrow$ state to $\downarrow$, and S^- flips $\downarrow$ to $\uparrow$. Using operators, the Hamiltonian (1.1.1) can be re-expressed as

$$H = \frac{J_\perp}{2}\sum_i \left(S_i^+ S_{i+1}^- + S_i^- S_{i+1}^+\right) + J_z \sum_i S_i^z S_{i+1}^z\,. \tag{1.1.9}$$

regarding S_i^z as a 'coordinate', $S_i^\pm$ displace this coordinate as if it :he 'momentum'. In this interpretation, the unitary operator U_i (1.1.3) ıponds to the linear transformation operator $U = e^{\mathrm{i}\alpha p}$ (x: coordinate; ımentum; α: constant). Then, the term proportional to $J_\perp$ in (1.1.9) ıents the 'kinetic energy' causing the quantum fluctuations of S_i^z, and

the term proportional to J_z represents the 'potential energy' that causes the ordering of S_i^z. The competition between these two tendencies is the physics that is contained in (1.1.1) as well as in (1.1.9).

We first consider the classical limit $J_\perp = 0$. This is the so-called Ising-model; and the spins align at zero temperature depending on the sign of J_z ferromagnetic ally ($J_z < 0$), or anti-ferromagnetically ($J_z > 0$). Notice that the ground state is two-fold degenerate because the Hamiltonian is invariant under the transformation $S_z^i \to -S_z^i$, performed at all sites i. Calling these two ground states A and B and assuming that the system at the right-hand side is in state A, and at the left-hand side in state B, then somewhere there must exist a boundary between region A and region B. This boundary is called a kink or soliton. Because at finite temperature this excitation occurs with a finite density, the spin correlation function $F(r) = \langle S_i^z S_{i+r}^z \rangle$ will decay exponentially with a correlation length ξ.

Let us determine this explicitly. We align N spins from $i = 1$ to $i = N$ with free ends. In the thermodynamic limit $N \to \infty$, we can ignore the influence of the boundary. We first assume that the spin S_1^z at the left side is fixed to, say, 1/2. Then, instead of S_2^z, it is possible to consider $L_{12} \equiv S_1^z S_2^z$ as variable. In the same manner, instead of S_3^z, defining $L_{23} \equiv S_2^z S_3^z$ and so on, the statistical mechanics can be formulated in terms of $L_{i,i+1}$ instead of S_i^z. The correlation between S_i^z and S_{i+1}^z, being defined on the site, is expressed by $L_{i,i+1}$, being defined on the link. In the parallel or anti-parallel case, the obtained value is $\pm 1/4$, respectively. Now, we write $L_{i+1/2}$ instead of $L_{i,i+1}$. Then, the Ising model can be expressed as

$$H = J_z \sum_{i=1}^{N-1} L_{i+1/2}\,, \tag{1.1.10}$$

which is the independent sum of the energy at every link. The free energy of every site can be calculated easily:

$$\begin{aligned} f &= -\frac{1}{\beta}\lim_{N\to\infty}\frac{1}{N}\ln Z = -\frac{1}{\beta}\ln\sum_{L=\pm 1/4} \mathrm{e}^{-\beta J_z L} \\ &= -\frac{1}{\beta}\ln\left(2\cosh\frac{\beta J_z}{4}\right). \end{aligned} \tag{1.1.11}$$

Next, based on the above discussion, we calculate $F(r) = \langle S_i^z S_{i+r}^z \rangle$. First, we observe that $S_i^z S_{i+r}^z$ is 'non-local' when expressed in terms of $L_{j+1/2}$. That is,

$$4S_i^z S_{i+r}^z = (4L_{i+1/2})(4L_{i+3/2})\ldots(4L_{i+r-1/2}) \tag{1.1.12}$$

can be expressed as the product of r terms in L connecting the sites i and $i+r$. With (1.1.10), from this fact we obtain

$$4F(r) = \prod_{j=1}^{r} \left[\frac{\sum_{L=\pm 1/4} 4L\, \mathrm{e}^{-\beta J_z L}}{\sum_{L=\pm 1/4} \mathrm{e}^{-\beta J_z L}} \right]$$
$$= \left[\tanh\left(\frac{-\beta J_z}{4} \right) \right]^r$$
$$= (-\operatorname{sgn} J_z)^r \exp\left[r \ln \tanh\left(\frac{\beta |J_z|}{4} \right) \right] . \qquad (1.1.13)$$

ing the absolute value of the left-hand side of (1.1.13) as $\mathrm{e}^{-r/\xi}$, the lation length ξ is determined to be

$$\xi = -\frac{1}{\ln \tanh\left(\frac{\beta |J_z|}{4} \right)} . \qquad (1.1.14)$$

ıermore, at low temperature $\beta |J_z| \gg 1$, in the limit where a small num- f kinks are thermally excited, equation (1.1.14) becomes

$$\xi^{-1} = 2\, \mathrm{e}^{-\beta |J_z|/2} . \qquad (1.1.15)$$

ling that the creation energy for a kink is given by $\Delta E = |J_z|/2$, the hand side corresponds to the number of kinks, and ξ can be interpreted ; mean distance between two kinks. That is, the long-range order $F(r) =$ gn $J_z)^r$ at $T = 0\,\mathrm{K}$ is destroyed owing to kink excitation. Notice that a :omes up as a change in sign of $L_{i+1/2}$.

ıe classical statistical mechanics described above almost did not depend e sign of J_z. The only dependence emerges in $(-\operatorname{sgn} J_z)^r$ in (1.1.13), ndeed, J_z changes to $-J_z$ when in the Hamiltonian all S_i^z are altered $_i^z$. When proceeding to quantum mechanics, this is no longer correct. s owing to the following reason. The term proportional to $J_\perp$ in (1.1.9) ıg the quantum mechanical motion expresses the simultaneous flip of eighbouring spins. It can be expressed as a product of S^+ and S^-, ore when one S^z is growing, the other must diminish, and the sum S_{i+1}^z is conserved. This simultaneous spin flip is the change between $1/2, S_{i+1}^z = -1/2\rangle$ and $|S_i^z = -1/2, S_{i+1}^z = 1/2\rangle$. For parallel spins, this e cannot occur. In a more mathematical language, let us define the total perator:

$$\boldsymbol{S}_{\mathrm{tot}} = \sum_i \boldsymbol{S}_i . \qquad (1.1.16)$$

$S_{\mathrm{tot}}^2 = \boldsymbol{S}_{\mathrm{tot}} \cdot \boldsymbol{S}_{\mathrm{tot}}$, and S_{tot}^z commute with the Hamiltonian (1.1.9):

$$[S_{\mathrm{tot}}^2, H] = [S_{\mathrm{tot}}^z, H] = 0 . \qquad (1.1.17)$$

s, the quantum system can be decomposed into eigenstates of S_{tot}^2 and n the case when $J_z < 0$ of the ferromagnetic interaction, it is clear

that the classical ground states $|\{S_i^z = 1/2\}\rangle$ and $|\{S_i^z = -1/2\}\rangle$ are also the eigenstates of the Hamilton operator H. On the other hand, the classical ground state in the anti-ferromagnetic case (the so-called Néel state) is not an eigenstate of the Hamiltonian. This becomes clear by constructing the staggered magnetization

$$\boldsymbol{S}_{\mathrm{staggered}} = \sum_i (-1)^i \boldsymbol{S}_i . \qquad (1.1.18)$$

The Néel state is an eigenstate of $S_{\mathrm{staggered}}^z$. However, $[S_{\mathrm{staggered}}^z, H]$ is different from zero. That is, for $S_{\mathrm{staggered}}^z$ there must exist a zero point quantum fluctuation. The physical picture for this zero point fluctuation is just the resonance between the states $|S_i^z = 1/2, S_{i+1}^z = -1/2\rangle$ and $|S_i^z = -1/2, S_{i+1}^z = 1/2\rangle$ that has been mentioned before. Considering for simplicity only two spins, $\boldsymbol{S}_i$ and $\boldsymbol{S}_{i+1}$, setting $J_\perp > 0$, owing to this resonance the ground state becomes a linear combination of two states

$$|\mathrm{singlet}\rangle = \frac{1}{\sqrt{2}} \left(\left| \frac{1}{2}, -\frac{1}{2} \right\rangle - \left| -\frac{1}{2}, \frac{1}{2} \right\rangle \right) . \qquad (1.1.19)$$

This is the spin singlet wave function. That is, the quantum fluctuation of S_i^z leads to the singlet formation. Equation (1.1.9) can be interpreted as competition between the tendency of $J_\perp$ to create singlet states leading to the emergence of a quantum liquid, and the tendency of J_z to order the spins.

Above, considerations concerning the ground state have been done. Next, we consider the first excited state based on the Néel ground state of the classical system. This is the domain wall as shown in Fig. 1.1. In what follows, we think about the Ising limit $J_z \gg J_\perp > 0$, where mixing between states with different numbers of domain walls can be ignored. First, we consider the state containing one domain wall. Calling Ψ_n the wave function of the state where a domain wall is present between site n and $n+1$, we obtain

$$(H - E_{\text{Néel}})\Psi_n = \frac{J_z}{2}\Psi_n + \frac{J_\perp}{2}(\Psi_{n+2} + \Psi_{n-2}) . \qquad (1.1.20)$$

Constructing a plane wave state of a domain wall

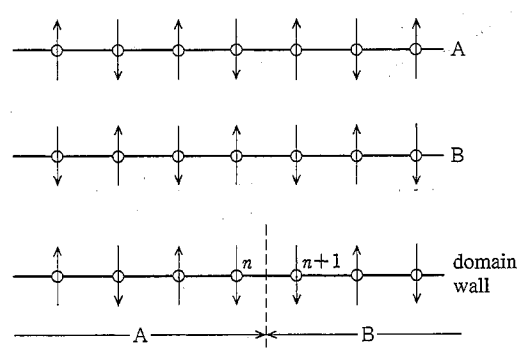

Fig. 1.1. The two degenerate classical Néel ground states A and B and the domain wall

$$\Psi(k) = \frac{1}{\sqrt{N}} \sum_n \mathrm{e}^{\mathrm{i}kn} \Psi_n \,, \tag{1.1.21}$$

btain

$$(H - E_{\text{Néel}})\Psi(k) = \left(\frac{J_z}{2} + J_\perp \cos 2k\right) \Psi(k) \,. \tag{1.1.22}$$

efore, the excitation energy is given by

$$\varepsilon_{\text{DW}}(k) = \frac{J_z}{2} + J_\perp \cos 2k \,. \tag{1.1.23}$$

n one domain wall is present (in general, an odd number), the sign of aggered magnetization of the configuration at the boundaries $n = -\infty$ $= +\infty$ changes. Therefore, no long-range order is present in the Ising l at finite temperature as discussed above. Here, because we consider d states at zero temperature, we set the staggered magnetization at $\pm\infty$ to be equal, corresponding to periodic boundary condition. Then, st two domain walls have to be created. Calling the momenta k_1 and k_2, ctively, the total momentum q and the excitation energy ΔE are given

$$\begin{aligned} q &= k_1 + k_2 \\ \Delta E &= \varepsilon_{\text{DW}}(k_1) + \varepsilon_{\text{DW}}(k_2) \,. \end{aligned} \tag{1.1.24}$$

ossible values for ΔE for every q are given by the oblique region in .2. In particular, the lower boundary is given by

$$\Delta E_{\text{LB}} = J_z - 2J_\perp |\cos q| \,. \tag{1.1.25}$$

nclude that the excitation spectrum does not consist of isolated spin states, but is the continuum, reflecting the fact that the elementary tions are kinks (domain walls).

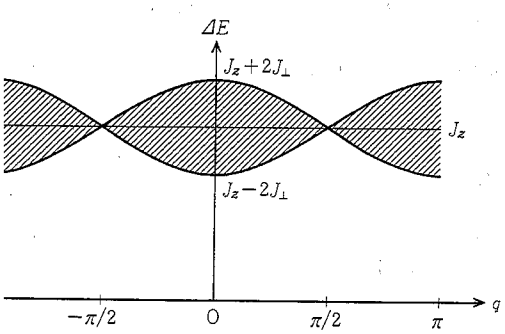

Fig. 1.2. The energy of the domain wall pair

1.2 The Jordan–Wigner Transformation and the Quantum Kink

In the previous section, the one-dimensional Ising model has been discussed as an example to demonstrate the importance of the kink. In this chapter, the quantum mechanics of the kink will be discussed in more detail. Starting with the conclusion, we note that the kink can be described in terms of fermions obtained by the following so-called Jordan–Wigner transformation

$$S_i^+ = S_i^x + \mathrm{i}S_i^y = f_i^\dagger K(i) = K(i) f_i^\dagger \,, \tag{1.2.1a}$$

$$S_i^- = S_i^x - \mathrm{i}S_i^y = K(i) f_i = f_i K(i) \,, \tag{1.2.1b}$$

$$S_i^z = f_i^\dagger f_i - \tfrac{1}{2} \,. \tag{1.2.1c}$$

Here, $K(i)$ is a non-local operator, defined by

$$K(i) = \exp\left[\mathrm{i}\pi \sum_{j=1}^{i-1} f_j^\dagger f_j\right] = \exp\left[\mathrm{i}\pi \sum_{j=1}^{i-1} \left(S_j^z + \frac{1}{2}\right)\right] \,. \tag{1.2.2}$$

$K(i) = K(i)^\dagger$, and $[K(i), K(j)] = 0$. Furthermore, for $i \le j$, $K(i)$ and $\boldsymbol{S}_j$ also commute. From (1.2.1) we conclude:

$$\begin{aligned} f_i^\dagger &= S_i^+ K(i) \,, \\ f_i &= S_i^- K(i) \,. \end{aligned} \tag{1.2.3}$$

From these equations, the anti-commutation relation of fermions can be deduced.

We first examine the meaning of $K(i)$. Using the unitary operator U_i introduced in (1.1.3), we can write

$$K(i) = \left[\prod_{j=1}^{i-1} U_j(\boldsymbol{e}_z, \pi)\right] \mathrm{e}^{\frac{\mathrm{i}\pi}{2}(i-1)} \,. \tag{1.2.4}$$

We conclude that $K(i)$ rotates all spins from $j = 1$ to $i - 1$ by π around the z axis; that is, this operator shifts S^x and S^y to $-S^x$ and $-S^y$. $K(i)$ is a non-local spin rotating operator, and therefore creates a kink. Now, for $i < j$ we consider

$$f_i f_j = S_i^- K(i) S_j^- K(j) \,. \tag{1.2.5}$$

From the above considerations, we obtain

$$K^\dagger(j) S_i^- K(j) = -S_i^- \tag{1.2.6}$$

that is, $K(j)$ and S_i^- do anti-commute (for $j > i$). Finally, the right-hand side of (1.2.5) becomes

$$S_i^- K(i) S_j^- K(j) = S_i^- S_j^- K(i)K(j) = S_j^- S_i^- K(i)K(j)$$
$$= -S_j^- K(j) S_i^- K(i) = -f_j f_i \tag{1.2.7}$$

therefore

$$\{f_i, f_j\}_+ = 0 \tag{1.2.8}$$

s. In the same manner, for $i < j$, because of

$$f_i f_j^\dagger + f_j^\dagger f_i = S_i^- K(i) S_j^+ K(j) + S_j^+ K(j) S_i^- K(i)$$
$$= S_i^- S_j^+ K(i)K(j) - S_j^+ S_i^- K(j)K(i)$$
$$= \left[S_i^-, S_j^+\right] K(i)K(j) = 0\,,$$

btain

$$\{f_i, f_j^+\} = \delta_{ij}\,. \tag{1.2.9}$$

iis manner, the non-local spin rotating operator (in the xy-plane) trans- the commutation relation of the spin operators at different sites into nti-commutation relation of fermion operators. As follows from (1.2.1c), ie site, the two possible states $S_i^z = \pm 1/2$ correspond to fermion num- $n_i = f_i^\dagger f_i = 1, 0$, respectively. We see how the Pauli principle naturally ibes the limited possibilities for the allowed spin states. That these ions correspond to kinks can be seen by considering, for example, the $|F\rangle$ where all spins are in the state $S_i^x = +1/2$. Acting with $f_n^\dagger$, we n the state $f_n^\dagger |F\rangle$ with $S_i^x = -\frac{1}{2}$ $(1 \le i < n)$, $S_n^x = +\frac{1}{2}$, $S_i^z = +\frac{1}{2}$ n).

he Jordan–Wigner transformation works well because although (1.2.1) n-local, the Hamiltonian (1.1.1) [(1.1.9)] can be expressed with these ons in a local manner. First, we consider the term proportional to $J_\perp$:

$$S_i^+ S_{i+1}^- = f_i^+ K(i) K(i+1) f_{i+1}$$
$$= f_i^+ \exp\left[\mathrm{i}\pi f_i^\dagger f_i\right] f_{i+1} = f_i^+ f_{i+1}\,. \tag{1.2.10}$$

we used the fact that in the state multiplied with $\exp[\mathrm{i}\pi f_i^\dagger f_i]$, owing to the ith site no more fermion can be present. Also, the term proportional can be expressed locally using (1.2.1c), leading finally to

$$= -\frac{J_\perp}{2} \sum_{i=1}^{N} \left(f_i^\dagger f_{i+1} + f_{i+1}^\dagger f_i\right) + J_z \sum_{i=1}^{N} \left(f_i^\dagger f_i - \frac{1}{2}\right)\left(f_{i+1}^\dagger f_{i+1} - \frac{1}{2}\right). \tag{1.2.11}$$

entioned in Sect. 1.1, the sign of $J_\perp$ can be chosen freely, therefore we ut a minus sign before $J_\perp > 0$. We obtained the spinless fermion model nearest site interaction J_z.

t this point, we discuss the periodic boundary conditions. Considering and setting

$$\boldsymbol{S}_{N+1} = \boldsymbol{S}_1\,, \tag{1.2.12}$$

the term $i = N$ proportional to $J_\perp$ in (1.2.11) becomes

$$\frac{J_\perp}{2}\left(S_N^+ S_1^- + S_N^- S_1^+\right). \tag{1.2.13}$$

Here, we obtain

$$S_N^+ S_1^- = K(N) f_N^\dagger f_1 = -K f_N^\dagger f_1\,. \tag{1.2.14}$$

The factor K can be expressed as

$$K = \exp\left[\mathrm{i}\pi \sum_{i=1}^{N} f_i^\dagger f_i\right] = (-1)^M \tag{1.2.15}$$

using the total number $M = \sum_{i=1}^{N} f_i^\dagger f_i$ of fermions. The minus sign on the right-hand side of (1.2.14) arises because in $K(N)$, $f_N^\dagger f_N$ is not contained. Defining

$$f_{N+1} = -f_1 \qquad (M : \text{even}) \tag{1.2.16a}$$
$$f_{N+1} = f_1 \qquad (M : \text{odd}) \tag{1.2.16b}$$

then (1.2.11) is valid as it stands.

As was mentioned in the previous section, (1.2.11) expresses competition between the formation of a spin singlet state and magnetic ordering, that is, competition between the kinetic energy of the fermions, i.e. itineracy, and the formation of density waves owing to the particle interaction. One remarkable feature of (1.2.11) is that in the case $J_z = 0$ (XY model); that is, in the quantum limit, the model becomes a free fermion theory that can be solved exactly. This limit is the opposite to the Ising limit that has been mentioned in the previous section. Introducing the Fourier transformations

$$f_n = \frac{1}{\sqrt{N}} \sum_k f_k\, \mathrm{e}^{\mathrm{i}kn}\,, \tag{1.2.17a}$$

$$f_n^\dagger = \frac{1}{\sqrt{N}} \sum_k f_k^\dagger\, \mathrm{e}^{-\mathrm{i}kn}\,, \tag{1.2.17b}$$

we obtain from (1.2.11)

$$H_{XY} = \sum_k \varepsilon(k) \cdot f_k^\dagger f_k \qquad (\varepsilon(k) = -J_\perp \cos k)\,. \tag{1.2.18}$$

The energy dispersion is shown in Fig. 1.3.

Owing to (1.2.1c), the relationship between the fermion number M and S_{tot}^z is $S_{\text{tot}}^z = M - N/2$. The Hilbertspace $S_{\text{tot}}^z = 0$ corresponds to the half-filled case $M = N/2$. In this case, the ground state is given by the state where the fermions occupy all states up to the Fermi energy $E_{\text{F}} = 0$. Excited states can be expressed as particle–hole creation. When N is even, the ground state is a singlet $S_{\text{tot}} = 0$, and the excited states start with $S_{\text{tot}} = 0$ or $S_{\text{tot}} = 1$.

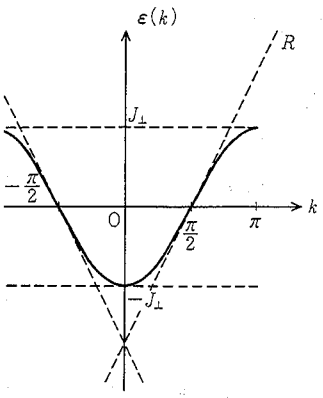

Fig. 1.3. The dispersion $\varepsilon(k) = J_\perp \cos k$. The dotted lines represent the linear approximation of the dispersion relation

fact that this is expressed as particle–hole creation shows that fermions spin $S = 1/2$.

oughly speaking, one spin flip is realized by introducing two kinks, and fore two fermions are necessary. In Fig. 1.4, the excitation energy of the cle–hole pair is shown as a function of $q = k_p - k_h$ (with k_p and k_h being xcitation energy of the particle and the hole, respectively).

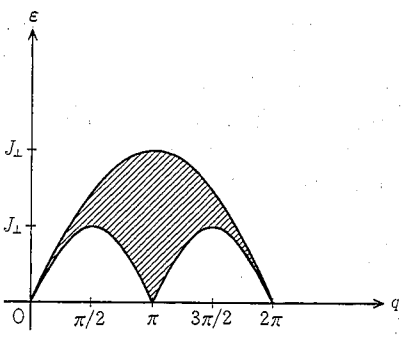

Fig. 1.4. The excitation of a particle–hole pair with wave number q

such a manner, in the XY model, it is not the spin itself, but rather ink – in the present case expressed as the Jordan–Wigner fermion – is the elementary excitation, and the spin excitations can be expressed ms of such combinations. In interacting particle systems – called many- systems – in many cases new particles different from those that describe riginal model are more fundamental. The quantum kink described here h an example.

1.3 The Bethe Ansatz and the Exact Solution

In Sect. 1.1 and 1.2, we examined the classical limit ($J_\perp = 0$) and the quantum limit ($J_z = 0$) of the Hamiltonian (1.1.1). In fact, there exists an exact solution of the Hamiltonian (1.1.1) [(1.1.9)]. However, even when the wave function of the ground state is determined, this does not mean that all physical quantities can be calculated. Only by combining various theoretical approaches will it be possible to obtain a complete picture.

First, we start with the simple case of ferromagnetic interaction ($J_z < 0$). We indicate n_i as the site with spin $\uparrow$ at the ith side from the left. The wave function of the state where r spins are $\uparrow$ can be written as

$$\Phi_{n_1 n_2 \ldots n_r} \,. \tag{1.3.1}$$

The total number of spins is given by N, therefore $N - r$ spins are $\downarrow$. The state Φ_0 where all spins are $\downarrow$ is an eigenfunction of (1.1.1):

$$H\Phi_0 = \frac{J_z}{4} N \Phi_0 \equiv E_0 \Phi_0 \,. \tag{1.3.2}$$

Here, we assume that the N spins are aligned as a one-dimensional ring (fulfilling periodic boundary conditions). Next, we consider the space of wave functions where just one spin is $\uparrow$, and with c_n being the coefficients of the linear superposition, we write

$$\Psi = \sum_{n=1}^{N} c_n \Phi_n \qquad (c_{n+N} = c_n) \,. \tag{1.3.3}$$

Acting with (1.1.1) on this state, we obtain

$$\begin{aligned} H\Psi &= \frac{J_z}{4}(N-4)\Psi + \frac{J_\perp}{2} \sum_{n=1}^{N} c_n (\Phi_{n-1} + \Phi_{n+1}) \\ &= \frac{J_z}{4}(N-4)\Psi + \frac{J_\perp}{2} \sum_{n=1}^{N} (c_{n+1} + c_{n-1}) \Phi_n \,. \end{aligned} \tag{1.3.4}$$

Writing $E\Psi$ for the right-hand side of (1.3.4) and comparing the coefficients of the Φ_n, we obtain

$$E c_n = (E_0 - J_z) c_n + \frac{J_\perp}{2}(c_{n+1} + c_{n-1}) \,. \tag{1.3.5}$$

This equation can be solved with plane waves

$$c_n = \frac{1}{\sqrt{N}} \mathrm{e}^{\mathrm{i}kn} \,, \tag{1.3.6}$$

leading to the eigenvalues